

1. Log Analysis Practice

Windows Event Viewer – Security Logs

Objective: Detect failed login attempts and suspicious service creation.

Steps:

1. Open **Event Viewer**

- Win + R → `eventvwr.msc`

2. Navigate to:

Windows Logs → Security

3. Click **Filter Current Log**

4. Use these Event IDs:

- **4625** → Failed login attempts
- **7045** → New service installed (possible persistence)

Brute-Force Identification:

- Look for:

- Same **Source IP**
- Multiple **4625** events

- Short time window
- Indicators:
 - Logon Type 3 (network)
 - Status **0xC000006A** (wrong password)

Export Evidence:

- Right-click → **Save Filtered Log As**
- Format: **CSV**
- Use for reporting and dashboards

Browser History Analysis (Chrome):

Objective: Identify suspicious or malicious URLs.

Tools:

- **LECcmd** (Eric Zimmerman's tool)

Steps:

1. Copy Chrome History file:

C:\Users\<user>\AppData\Local\Google\Chrome\User
Data\Default\History

2. Run:

LECcmd.exe -f History --csv output

3. Open CSV → Filter for:

- <http://test.com>
- Unknown domains
- High visit frequency

Use Case: Detect phishing, C2 callbacks, or malicious browsing.

2. Document Security Events:

Incident Documentation Template

Use this table format:

Date & Time	Source IP	Event ID	Description	Action Taken
2026-03-02 10:32	192.168.100.16	4625	Multiple failed logins	IP flagged, monitoring enabled

Best Practices:

- Be clear and factual
- Avoid assumptions
- Mention **next steps** (monitor, block, escalate)

This is exactly how **Tier-1 SOC analysts** document alerts.

3. Set Up Monitoring Dashboards:

Dashboards in Kibana / Grafana

Visualizations to Create:

- Top 10 source IPs generating alerts

- Event frequency over time
- Count of critical Event IDs (4625, 7045)

Data Sources:

- Elastic indices (e.g., `security-login-*`)
- Wazuh alerts index

Bonus:

- Import **Sigma detection rules**
- Convert Sigma → Elastic rules

Tools commonly used:

- Elastic SIEM
- Kibana
- Grafana

4. Configure Alert Rules:

Elastic SIEM Rule – Failed Login Detection

Rule Logic:

- Condition: `5+ failed logins`
- Time window: `5 minutes`
- Index: `security-login-*`

Test:

- Generate failed logins on Windows or Linux
- Confirm alert fires
- Document:
 - Detection time
 - Accuracy
 - False positives

Advanced: Wazuh Custom Rule

Goal: Detect brute-force SSH attempts.

Rule Logic:

- 3+ failed logins
- 2 minutes
- Source: SSH auth logs

Testing:

ssh user@192.168.100.16

Enter wrong password multiple times

Validation:

- An alert appears in **Wazuh dashboard**
- Record:
 - Alert severity
 - Trigger count

- Effectiveness

5. Learning & SOC Workflow Mapping

Mini-SOC Lab Setup

- SIEM: **Wazuh**
- Endpoint visibility: **Osquery**
- Logs: Windows + Linux

MITRE ATT&CK Mapping

Map alerts to tactics:

- **T1110** → Brute Force
- **T1059** → PowerShell execution
- **T1547** → Persistence via services

Use **MITRE ATT&CK** in your report to show analyst-level thinking.